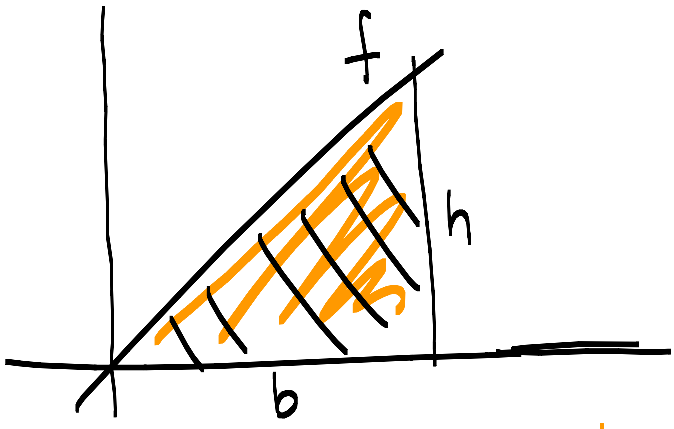
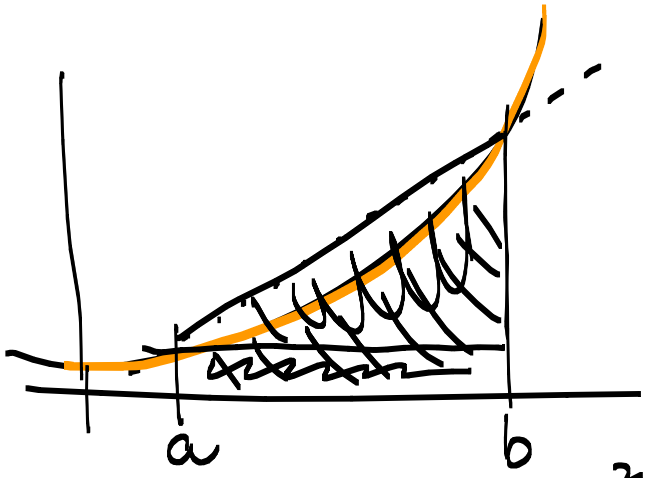
Note Title 2014/04/20

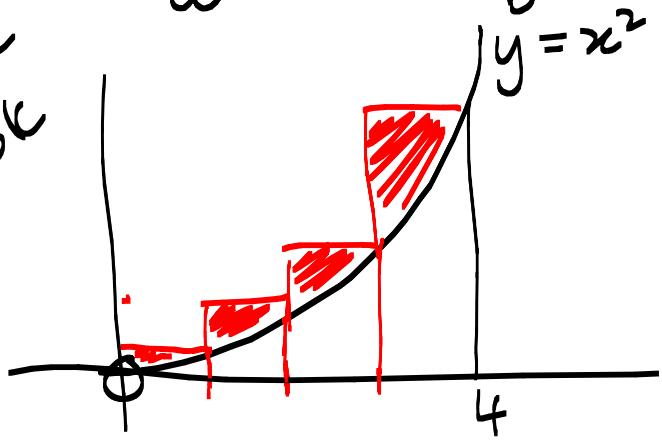
Unit 5.2 The area problem p 360 - 367

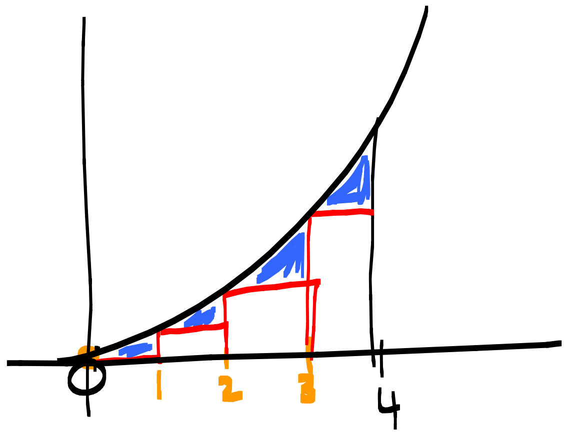


Area = $\frac{1}{2} b \times h$



Example in book





Example: $f(x) = x^2 + 4$ on $[-2, 4]$

using 3 rectangles

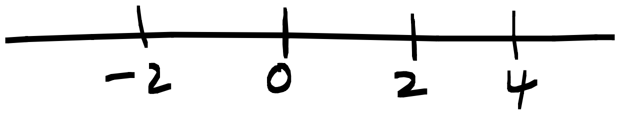
$$\Delta x = \frac{b-a}{n}$$ — endpoints

length

number of rect

$$\Delta x = \frac{4-(-2)}{3} = 2$$

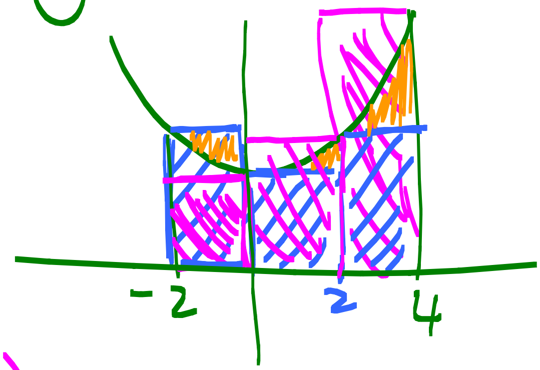
$f(x) = x^2 + 4$



(b) x	-2	0	2	4
(h) $f(x)$	8	4	8	20

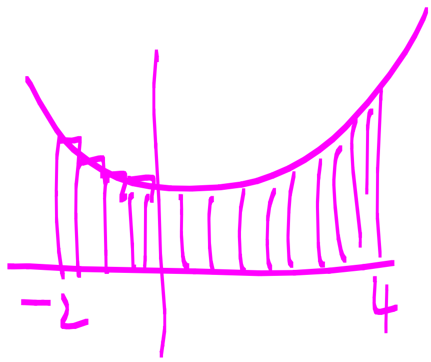
Area below the curve using Left Hand Sum

$$LHS = 2(8) + 2(4) + 2(8)$$
$$= 40$$



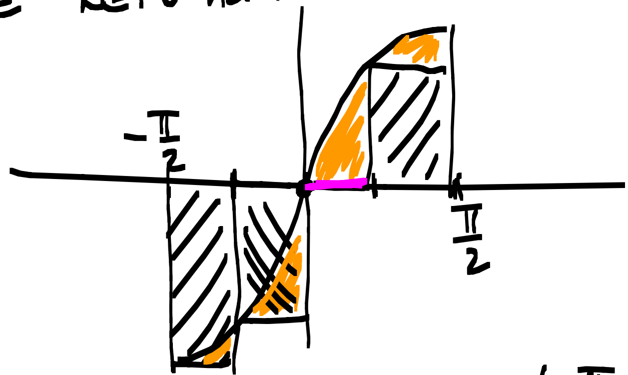
$$RHS = 2(4) + 2(8) + 2(20)$$
$$= 64$$

$$40 \le \text{area} \le 64$$



Example: Find the area under the curve of $f(x) = \sin x$ on $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ using 4 rectangles. and the left hand sum.

$$\Delta x = \frac{b-a}{n}$$
$$= \frac{\frac{\pi}{2} - \left(-\frac{\pi}{2}\right)}{4} = \frac{\pi}{4}$$



(b) x	$-\frac{\pi}{2}$	$-\frac{\pi}{4}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$
(h) $f(x)$	-1	$-\frac{1}{\sqrt{2}}$	0	$\frac{1}{\sqrt{2}}$	1

LHS

$$= \frac{\pi}{4}\left[1 + \frac{1}{\sqrt{2}} + 0 + \frac{1}{\sqrt{2}}\right] = \text{etc.}$$